DELIVERY ROUTE OPTIMIZER

Data Structures and Algorithms – 3

V SRI VAISHNAVI – 2520090117

MOHANA PRIYA – 2520090176

SECTION: 10

GUIDED BY MS. K. CHANDUSHA, ASST. PROF, CS&IT



ABSTRACT

A web-based application for efficient delivery routing

Locations modeled as nodes, roads as edges with distances

Dijkstra's Algorithm computes the shortest path

PriorityQueue (Min-Heap) speeds up nearest location selection

Outputs route distance, estimated time, and supports multiple deliveries



PROBLEM STATEMENT

The Challenge

MULTIPLE DELIVERY POINTS MAKE ROUTE PLANNING COMPLEX

MANUAL ROUTES INCREASE TIME, DISTANCE, FUEL, AND COST

HANDLING MANY DELIVERY REQUESTS IS INEFFICIENT

Our Solution

MODEL THE DELIVERY NETWORK AS A GRAPH

USE DIJKSTRA'S ALGORITHM TO FIND THE SHORTEST ROUTES

PROVIDE A WEB INTERFACE TO DISPLAY ROUTE DETAILS



TECHNOLOGIES & REQUIREMENTS

SOFTWARE

HTML, CSS, JAVASCRIPT

JAVA

MYSQL DATABASE



HARDWARE

LAPTOP/DESKTOP

MINIMUM 4 GB RAM

INTERNET AND MODERN BROWSER



METHODOLOGY

1. ADD DELIVERY LOCATIONS
2. BUILD GRAPH OF LOCATIONS AND ROADS
3. ASSIGN DISTANCES TO ROADS
4. RUN DIJKSTRA'S ALGORITHM
5. USE PRIORITY QUEUE (MIN-HEAP) TO PICK NEAREST NODE
6. SHOW SHORTEST ROUTE, DISTANCE, AND TIME

EXPECTED RESULT & CONCLUSION

Find shortest routes between locations

Reduce travel distance and time

Save fuel and cost

Manage multiple delivery requests easily

Demonstrates practical use of DSA concepts

Efficient route alanning for logistics

Future improvements: GPS navigation, real-time traffic updates



THANK YOU

Any Questions?

WE APPRECIATE YOUR TIME

AND LOOK FORWARD TO YOUR QUESTIONS.

